Lab 5
Machine Learning Challenge: Air Quality Prediction

1. Dataset Description and Goal

Air pollution is one of the most serious environmental challenges worldwide. The Air Quality Index (AQI) is a standardized indicator used to describe how clean or polluted the air is and what associated health effects might be of concern. In this dataset, each record represents air quality measurements taken in various cities across different countries. The data includes the overall AQI and sub-indexes for key pollutants such as CO, Ozone (O_3), NO_2, and PM2.5, along with their qualitative categories (Good, Moderate, etc.).

Dataset Variables:

Variable	Description
Country, City	Geographic information about the measurement site.
AQI Value, AQI Category	Overall air quality indicator (numeric and categorical).
CO AQI Value, CO AQI Category	Carbon Monoxide pollution sub-index and qualitative label.
Ozone AQI Value, Ozone AQI Category	Ozone pollution sub-index and qualitative label.
NO_2 AQI Value, NO_2 AQI Category	Nitrogen Dioxide pollution sub-index and qualitative label.
PM2.5 AQI Value, PM2.5 AQI Category	Fine particulate matter sub-index and qualitative label.

Overall Goal: The objective of this challenge is to predict air quality levels using machine learning methods. You will apply regression models to predict the numeric AQI value and classification models to predict the AQI category. You will then compare the performance of different algorithms and interpret their results.

Let's start…

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2. Part 1 — Dataset Preprocessing

Goal: Prepare and explore the dataset to ensure it is ready for machine learning modeling. Add comments if needed to interpret your choices and results

Steps:

1. Data Understanding: Load and inspect the dataset (columns, types, missing values, duplicates, Multicollinearity, data profiling).
2. Data Cleaning: Handle missing or inconsistent values, remove duplicates, and check data types.
3. Encoding Categorical Variables **if needed**: Convert text categories into numeric or dummy variables.
4. Feature Selection and Normalization: Identify useful features and normalize numeric data if needed.
5. Data Splitting: Divide the dataset into training and testing subsets (e.g., 80/20).

3. Part 2 — Air Quality (AQI Value) Prediction (Regression Task)

Goal: Predict the overall AQI Value based on pollutant sub-index values and other relevant variables. Add comments if needed to interpret your choices and results

1. Variable Selection: Choose predictor variables and **justify** their relevance.
2. Model Training: Apply Linear Regression, Decision Tree Regressor, and Random Forest Regressor.
3. Model Evaluation: Use R^2, RMSE, and MAE to assess model performance and plot predicted vs actual AQI.
4. Results Interpretation: Analyze feature importance and discuss overfitting, bias, and variance issues.

4. Part 3 — Air Quality Category Prediction (Classification Task)

Goal: Predict the AQI Category using pollutant values.

1. Variable Selection: Identify relevant predictors and **justify** inclusion.
2. Model Training: Train Logistic Regression, Decision Tree, Random Forest, Perceptron, and MLP Classifier.
3. Model Evaluation: Use accuracy, precision, recall, F1-score, and confusion matrices to evaluate models.
4. Results Interpretation: Discuss performance differences, interpretability, and predictive power of models.

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5. Conclusion

In your conclusion, compare all the machine learning approaches used. Discuss which regression and classification models performed best, their strengths and limitations, and how model complexity influenced performance. Provide data-driven reflections and recommendations for real-world air quality prediction applications.